

Gurpreet (Gup) Singh

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Education

University of California, Berkeley

December 2022

Bachelor of Arts, Physics

Experience

Research Data Analyst I / Postgraduate Researcher, UC Berkeley Feb 2026-Present; Feb 2023-Aug 2024

- Adapted microcanonical Langevin Monte Carlo (MCLMC) to CUORE experimental likelihoods, with a custom implementation and parameter transformation, and developed a block variant; benchmarked against BAT.jl Metropolis-Hastings and HMC, including scaling with the number of datasets
- Built a Hamiltonian Monte Carlo sampler from scratch, with mass-matrix tuning, and a framework to validate samplers on posterior agreement, convergence, effective sample size, and runtimes
- Developed real-time signal processing and data acquisition software on AMD RFSoc hardware, achieving ~100 dB signal-to-noise ratio for low-noise experimental inference pipelines

Researcher, Leinweber Institute for Theoretical Physics

May 2023-Present

- Built a Bayesian parameter-inference pipeline for supersymmetric models (CMSSM, GSUGRA), combining renormalization-group running with experimental constraints
- Developed computational modeling and numerical methods that led to two published *Physical Review D* papers on chiral gauge theory dynamics

DOE SULI Internship, Lawrence Berkeley National Laboratory

Aug 2023-May 2024

- Processed 100M+ QCD simulated events using NERSC supercomputing GPU clusters, training classifiers with custom loss functions on large-scale datasets for particle physics anomaly detection
- Implemented and benchmarked PAWS, a weakly supervised anomaly detection method, improving sensitivity by 10× on the reported benchmark; the work resulted in a *Physical Review Letters* publication and an invited ML4Jets talk at Caltech

Research Assistant, UC Berkeley

Jan 2023-Jun 2023

- Built an end-to-end data processing workflow processing millions of particle collision events; deployed ML classifiers that distinguished charm-tagged baryons from background events

Projects

Founder, early-stage startup (May 2026 – Present)

- Prototyping a hardware/firmware product for commercial use (more information to be released)

Publications

Incorporating Physical Priors into Weakly Supervised Anomaly Detection – **Physical Review Letters** (ML/statistics) 2025

Generator Based Inference – **Data Science in Science** (ML/statistics) 2026

Exact Results in Softly Broken Supersymmetric Chiral Gauge Theories with Flavor – **Physical Review D** (Theoretical Physics) 2025

Dynamics of E_6 Chiral Gauge Theories – **Physical Review D** (Theoretical Physics) 2025

Presentations

ML4Jets, California Institute of Technology, Pasadena, CA

Aug 2025

Invited talk on our weakly supervised machine learning method for anomaly detection in particle physics

Skills

Languages: Python (NumPy, SciPy, PyTorch, TensorFlow), Java, C++/C, SQL, Julia, LaTeX, React, MATLAB

Compute & Tooling: NERSC GPU clusters, SLURM job orchestration, Git, LabVIEW

Hardware: Electrical Circuits, RF Signal Processing, Soldering, G-code

Honors & Awards

Berkeley Transfer Scholarship | The Colorado Space Grant Consortium | Osher Scholarship | Phi Theta Kappa Honors Society